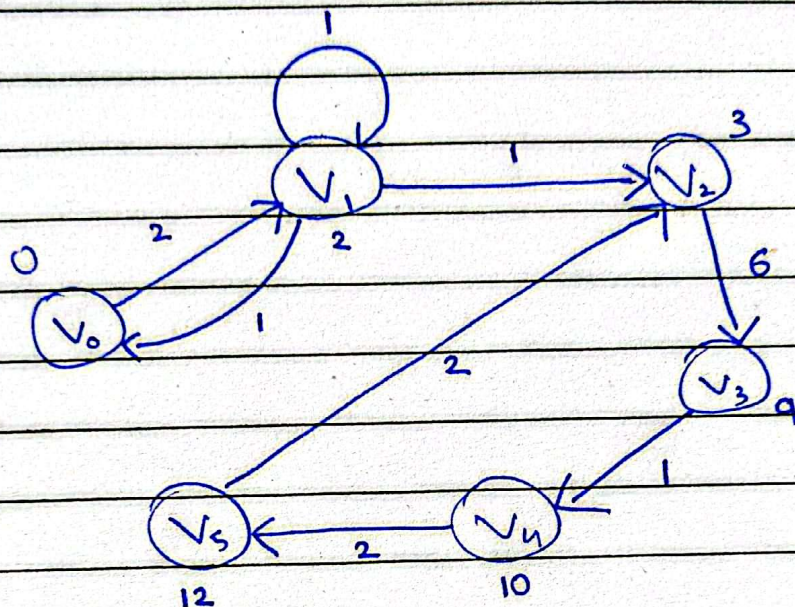


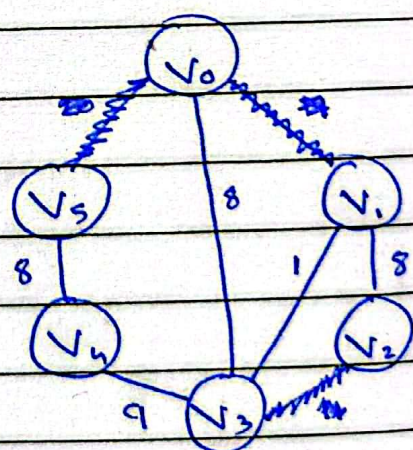
	V_0	V_1	V_2	V_3	V_4	V_5
V_0	0	1	0	0	0	0
V_1	1	0	1	0	0	0
V_2	0	0	0	1	0	0
V_3	0	0	0	0	1	0
V_4	0	0	0	0	0	1
V_5	0	0	1	0	0	0

	V_0	V_1	V_2	V_3	V_4	V_5
V_0	0	2	0	0	0	0
V_1	1	0	1	0	0	0
V_2	0	0	0	6	0	0
V_3	0	0	0	0	1	0
V_4	0	0	0	0	0	2
V_5	0	0	2	0	0	0

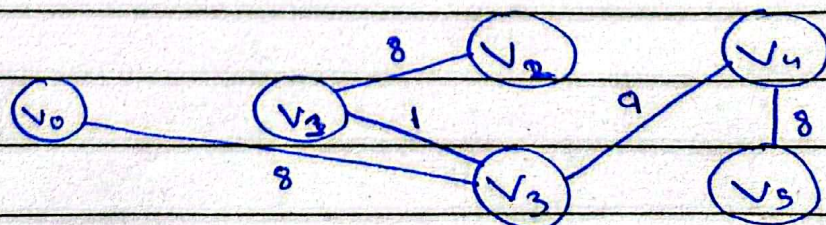


Using Floyd-Warshall algorithm, minimum cost is 12

	V_0	V_1	V_2	V_3	V_4	V_5
V_0	0	19	15	8	19	20
V_1	19	0	8	1	19	11
V_2	15	8	0	11	10	12
V_3	8	1	11	0	9	10
V_4	19	19	10	9	0	8
V_5	20	11	12	10	8	0



Clearer, using Kruskal's Algo.



Total five cables required for MST.

$$\text{Minimum installation cost} = 1 + 8 + 8 + 8 + 9 \Rightarrow 34$$

From 1st graph, SCC are $\{v_0, v_1\}$ & $\{v_2, v_3, v_4, v_5\}$

There is one link between the two SCCs, $v_1 \rightarrow v_2$

Calculation of matrix N ,

$$N = \begin{pmatrix} N(0,0) & N(0,1) \\ N(1,0) & N(1,1) \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$\therefore N(x,y)$ is the count of directed links from any node on cluster X_x to any node on cluster X_y .